# Shock in Office Practice

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## Introduction

Shock is one of the most common emergencies encountered in pediatric practice. Shock is defined as the inability of circulation to meet the metabolic demands of the body. Common types of shock are hypovolemic (dehydration/trauma), distributive (septic/anaphylactic), cardiogenic, and obstructive (i.e., pneumothorax and cardiac tamponade) with variable physiological derangements as outlined in **Table 1**.

**TABLE 1:** Physiological variables in different types of shock.

| *Type of shock* | *Preload* | *Afterload* | *Contractility* |
|---|---|---|---|
| Hypovolemic | ↓↓↓ | ↑ | ↑ |
| Distributive (septic/anaphylactic) | ↓ | ↓ or normal | ↓ or normal |
| Cardiogenic | ↑ | ↑ | ↓↓↓ |
| Obstructive | ↓ | ↑ | ↑ |

In office practice, pediatricians commonly encounter three types of shock scenarios:

1. *Acute gastroenteritis leading to hypovolemic shock*: Fluid therapy is the mainstay of treatment
2. *Septic shock*: Manage as per septic shock algorithm **(Fig. 1)**
3. *Anaphylactic shock*: Manage as per anaphylaxis algorithm **((Fig. 2)**

Sepsis is life-threatening organ dysfunction caused by a dysregulated host response to infection. Septic shock is a subset of sepsis with circulatory and cellular or metabolic dysfunction associated with a higher risk of mortality. In office practice, it is important to rapidly recognize shock based on early signs which may be subtle in the compensated stage. Without appropriate intervention, shock will progress from compensated to a hypotensive or irreversible stage **(Table 2)**. For successful management, one must follow the evaluate-identify-intervene (E-I-I) sequence as discussed in the Indian Academy of Pediatrics (IAP) advanced life support program.

Introduction

**TABLE 2:** Evaluate and identify the stages of shock.

| ***Stages*** | ***Physical examination findings*** |
|---|---|
| *Compensated shock*<br>Organ function is maintained | *Early signs of shock:* Tachycardia, poor pulses, prolonged (>2 seconds) capillary refill time (CRT), cold peripheries (cold shock), reduced urine output, anxious-irritable child, and generally associated with fast breathing but *blood pressure (BP) is normal*. Some children may have bounding pulses with flushed CRT and warm peripheries (warm shock). This stage is frequently missed in absence of proper evaluation |
| *Hypotensive shock*<br>End-organ dysfunction<br>Microvascular failure | Worsening trend of above clinical features such as tachycardia, CRT >3 second, cold-clammy skin, oliguria-anuria, dull or drowsy, tachypnea with increased work of breathing along with *low BP*. Hypotension is defined is systolic BP (SBP) <60 mm Hg in term neonates, <70 mm Hg in infants, < (70+ age in years × 2) in 1–10-year old children and <90 in children above 10 years of age. More than 20–25% acute blood loss or fall of 10 mm of SBP from observed level should be considered significant |
| *Irreversible shock and cardiac arrest*<br>End-organ cellular death | Bradypnea-apnea, bradycardia, very prolonged CRT (>6 seconds), anuria, coma, seizures, and low to nonrecordable BP. This stage may soon progress to cardiac arrest |

Management (Intervention)

One must understand three phases for management of shock, i.e., rapid recognition, stabilization/resuscitation, and further critical care management in the pediatric intensive care unit (PICU).

1. **Rapid recognition of shock (first 5 minutes):** It is crucial and based on a quick primary assessment as discussed under evaluation and identification **(Table 2)**. Never forget the clinical signs of altered end-organ perfusion, i.e., decreased urine output (<1 mL/kg/h), altered mental status (anxiety, restlessness, seizure, or loss of consciousness) and altered skin perfusion [flush or prolonged capillary refill time (CRT)]. Shock should be recognized and intervened in the compensated stage. Hypotension is a late sign and the child may rapidly progress to cardiac arrest after hypotension sets in.
2. **Initial stabilization and resuscitation:** In this phase, one must ensure increased oxygen delivery to tissues and reduce oxygen demand. While managing the child if anaphylaxis is suspected, switch to the anaphylaxis algorithm.

   *Within the first 10–15 minutes* of detection of signs of shock, airway, oxygenation, ventilation, and monitoring of heart rate/rhythm and pulse oximetry should be taken care of and vascular access should be established.

a. **Positioning:** Supine position or most comfortable position for the responsive child.

b. **Support airway and breathing:** Ensure effective oxygenation and ventilation, start high concentration of oxygen preferably by high flow device [nonrebreather mask (NRM)]. If the child is in respiratory failure, ensure mechanical ventilation, continuous saturation of peripheral oxygen ($SpO_2$) monitoring and venous blood gas might help, look for serum lactate level also.

c. **Vascular access:** Preferably two, large-bore cannulae, or go for an interosseous needle if intravenous (IV) cannulation is not possible. Central venous access is desirable but not mandatory for inotropic support in an emergency.

d. **Fluid therapy in acute gastroenteritis with dehydration:** Follow the WHO guidelines for acute gastroenteritis management for dehydration. Do not forget to replace ongoing losses and always monitor sodium, potassium, calcium, sugar, and urine output.

e. **Fluid therapy in septic shock:**

   i. Start fluid therapy within the first 5 minutes of identification of shock in the form of isotonic crystalloid solution as a 10–20 mL/kg bolus over 10–15 minutes. Reassess after every bolus and repeat fluid boluses if needed to restore BP and perfusion.

   ii. In the pediatric office setting, total bolus fluid up to 40 mL/kg may be administered over the first hour in hypotensive septic shock while starting maintenance fluids. Do not give a bolus of fluids in the absence of hypotension. Keep caution for pulmonary edema, especially in case of anemia and severe febrile illness.

   iii. If cardiogenic shock is suspected, consider a small fluid bolus (5–10 mL/kg) over 10–20 minutes and reassess. Suspicion of cardiogenic shock is high usually if: (1) heart rate is very high, disproportionate to clinical settings and (2) poor perfusion with hepatomegaly and respiratory distress.

   iv. Standard resuscitation fluid is isotonic crystalloids (0.9% saline or Ringer's lactate or buffered crystalloids). Consider albumin and other colloids only in case of albumin deficiency or large third spacing and blood products in case of visible or occult blood loss. Hydroxyethyl starches should not be used.

> Concerns regarding Ringer's lactate causing hyperkalemia and lactic acidosis are false. Ringer's lactate contains sodium lactate, but not lactic acid. Lactate is beneficial as it is converted into bicarbonate (in the liver), a base element that helps regulate the body's pH balance and avoid acidosis. Ringer's lactate does include a concentration of potassium 4 mEq/L, which is very less and this fluid is good in any scenario. Balanced crystalloids are now the preferred fluid of choice in the pediatric sepsis pathway.

f. **Glucose control:** Blood glucose ≤60 mg/dL is used to define hypoglycemia (beyond the neonatal period). Hypoglycemia should be identified rapidly and corrected immediately. IV dextrose may be administered as 25% dextrose (2 mL/kg), 10% dextrose (5 mL/kg) or 5% dextrose (10 mL/kg). A single bolus of 25% can be given through a peripheral line.

g. **Calcium and hypocalcemia:** Ionized hypocalcemia may impair cardiac performance and should be corrected as it is common in neonates and children with sepsis.

h. **Bicarbonate therapy:** Do not use bicarbonate therapy for the purpose of improving hemodynamics or reducing vasopressor requirements when treating hypoperfusion-induced lactic acidemia with pH ≥ 7.15.

*Within the first hour*: Ensure vasoactive drugs as indicated and give first dose of broad-spectrum antibiotic in septic shock.

*Vasoactive drugs*: An appropriate vasoactive drug infusion should be started if there is no response to fluid therapy:

i. Use epinephrine for cold shock, norepinephrine for warm shock, and dopamine as an alternative (in both conditions).

ii. Vasoactive agents can be given through the peripheral line in emergency setting. Try to shift to the central line as soon as possible (restrict peripheral line vasoactive agent use to <6 hours). Vasoactive agents when given through a peripheral line need to be diluted more.

iii. Use arterial BP in addition to bedside clinical signs to categorize septic shock in children as "warm" or "cold".

*Antibiotics*: Preferred to collect blood culture and ensure the first dose of broad-spectrum antibiotic is given.

**TABLE 3:** Monitor the shock index in the management of septic shock.

| | |
|---|---|
| *Shock index* | Heart rate (HR)/systolic blood pressure<br>1.2 for 4–6 years; 1 for 6–12 years; and 0.9 for >12 years<br>For normal healthy adults: 0.5–0.7 |

3. **Therapeutic endpoints of shock, ongoing care, and further critical care management:** By the end of the first hour, after stabilization, one should ensure that child is shifted from emergency room (ER) or office setting to PICU for hemodynamic monitoring, titration or addition of newer vasoactive drugs, and ongoing care even if the therapeutic endpoints of shock are already achieved **(Table 4)**.

**TABLE 4:** Therapeutic endpoints of shock resolution and ongoing care.

| ***Therapeutic endpoints of shock*** | ***Organ support to continue in pediatric intensive care unit (PICU)*** |
|---|---|
| ☑ Normal heart rate (HR) or declining from very high to normal<br>☑ Normal peripheral pulses and capillary refill time < 2 seconds and warm extremities<br>☑ Normal mental status/responsiveness<br>☑ Normal blood pressure<br>☑ Urine output > 1 mL/kg/h<br>☑ Improving serum lactate and metabolic acidosis | ☑ Infection control/source control<br>☑ Mechanical ventilation<br>☑ Renal replacement therapy<br>☑ Intracranial pressure management<br>☑ Blood transfusion<br>☑ Nutritional supplementation<br>☑ Management of hypo- or hyperglycemia, dyselectrolytemia, venous thromboembolism, disseminated intravascular coagulation (DIC), etc.<br>☑ Speciality consultations<br>☑ Discussion with family on goals of care and prognosis |

**In office practice:**

- ☑ Recognize shock early
- ☑ Initiate appropriate care plan for different types of shock
- ☑ Get IV/IO access
- ☑ Start fluids along with supportive measures
- ☑ Promptly refer to a facility equipped with PICU after initial management.

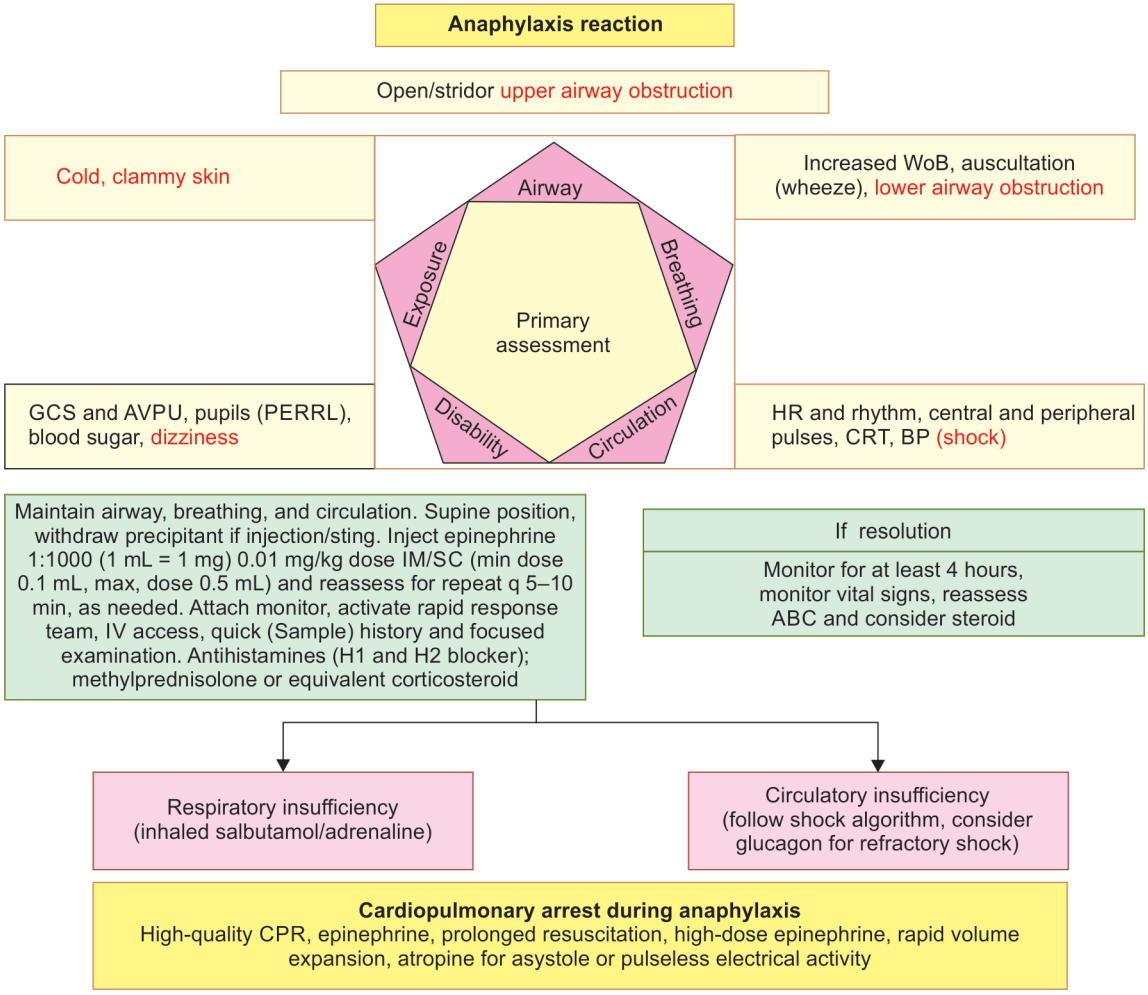


**Fig. 1:** Anaphylaxis algorithm. (ABC: airway, breathing, and circulation; AVPU: Alert Verbal Painful Unresponsiveness; BP: blood pressure; CPR: cardiopulmonary resuscitation; CRT: capillary refill time; GCS: Glasgow Coma Score; HR: heart rate; WoB: work of breathing)

## Management (Intervention)

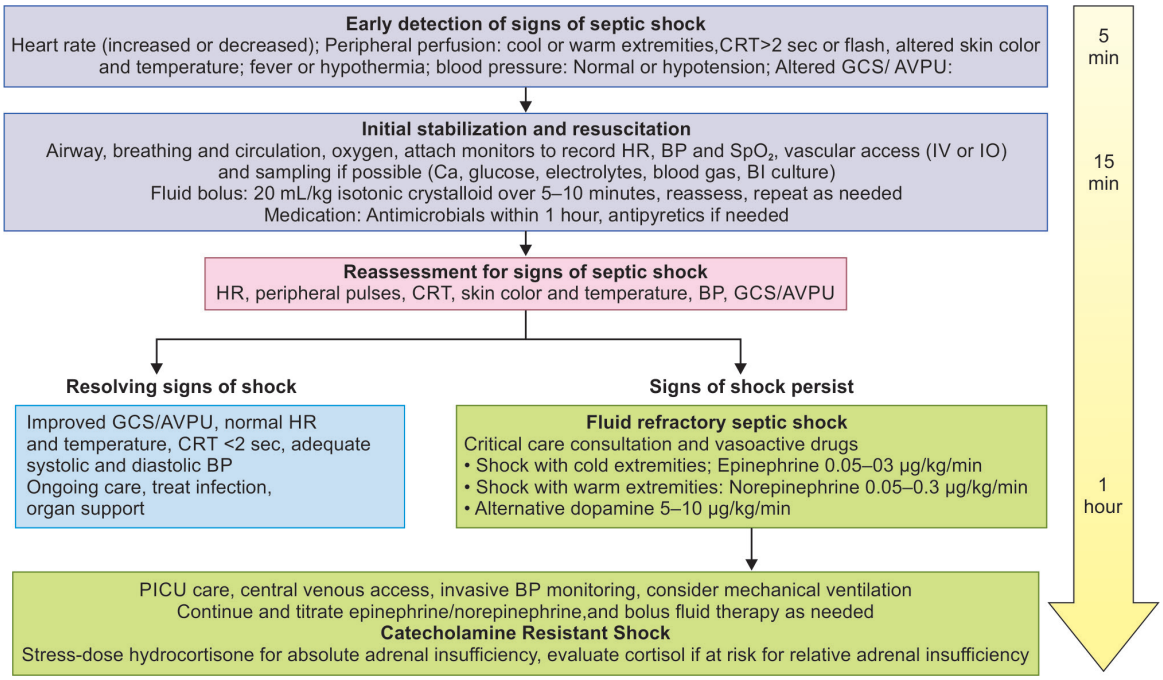


**Fig. 2:** Septic shock (stabilization before shifting to PICU) algorithm. (AVPU: Alert Verbal Painful Unresponsiveness; BP: blood pressure; CRT: capillary refill time; GCS: Glasgow Coma Score; HR: heart rate; PICU: pediatric intensive care unit; $SpO_2$: saturation of peripheral oxygen)

## Further Reading


- ☑ Rhodes A, Evans LE, Alhazzani W, Levy MM, Antonelli M, Ferrer R, et al. Surviving sepsis campaign: international guidelines for management of sepsis and septic shock: 2016. Intensive Care Med. 2017;43(3):304-77.
- ☑ Tiwari L, Chaturvedi J, Anand C. Myocardial dysfunction in sepsis. J Pediatr Crit Care. 2018;5:41-9.
- ☑ Weiss SL, Peters MJ, Alhazzani W, Agus MSD, Flori HR, Inwald DP, et al. Surviving sepsis campaign international guidelines for the management of septic shock and sepsis-associated organ dysfunction in children. Pediatr Crit Care Med. 2020;21(2):e52-e106.
- ☑ World Health Organization. Updated guideline: pediatric emergency triage, assessment and treatment. Geneva: World Health Organization; 2016.

Indian Academy of Pediatrics (IAP)

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# Respiratory Distress in the Term Newborn

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